

Rudra Gohil

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PROJECTS

Monk - A Programming Language From Scratch

- Monk is implemented as a **tree-walking style interpreter**. Tokenization , **Pratt-Parsing** and Evaluation is performed inside the REPL **without using any external Libraries**.
- It also has **functions, conditionals, implicit and explicit returns, recursive functions** and who doesn't love **closures** !

A Raytracer From The Streets

- It was a Mosaic of Programming and Math , Ranging from **Recursion for Reflections** , Object Oriented Programming to glue different Light Sources all the way to realm of **Linear algebra** and **sphere geometry** with numpy.
- Shadow Coherence was taken care of to **increase the efficiency** and Quaternions were used to **implement the rotation** and some **multiprocessing** at the end saved the day.

Girgit - Learn Git by Making one

- Following the Scratch Trilogy is girgit - a git like version control system **resembling a big linked-list** separating each little nodes with **SHA256**.
- A clear separation was made between the hash object(data) affecting functions , higher level data structures that use those functions and Command line Interface functionalities to ensure tidyness. The crown jewel was native implementation of **Myer's Algorithm** for diffing.
- Also a command with usage similar to **gitk tool** was implemented to **visualize the DAG tree** along with other standard commands.

DinoPDF - a game that lives inside pdf

open this pdf in chrome to play the game...

- The Portable Document Format is widely used to share **static documents** but it also provides **PDF Javascript API** along with formfields (**Acroforms**). A lot of tweaking and creative workarounds resulted in this game on the **next page**.

EDUCATION

Birla Vishvakarma Mahavidyalaya

Bachelor of Technology in Information Technology

8.86 CPI

Jul. 2023 – Jul 2027

Anand, Gujarat

Kendriya Vidyalaya

2019 – 2023

Anand, Gujarat

- JEE Mains 95.02 %ile
- Grade 12 92.2 %
- Grade 10 93.4 %

SKILLS

Languages : Python , Go(Beginner), MySQL

Tools : Manim, Git, GitHub, WireShark , Math